Evaluation

1. An allohexaploidy contains
   a) Six different genomes
   b) Six copies of three different genomes
   c) Two copies of three different genomes
   d) Six copies of one genome

2. Match list I with list II

List I	List II
A. A pair of chromosomes extra with diploid	i) monosomy
B. One chromosome extra to the diploid	ii) tetrasomy
C. One chromosome loses from diploid	iii) trisomy
D. Two individual chromosomes lose from diploid	iv) double monosomy

a) A-i, B-iii, C-ii, D-iv b) A-ii, B-iii, C-iv, D-i
c) A-ii, B-iii, C-i, D-iv d) A-iii, B-ii, C-i, D-iv

3. Which of the following sentences are correct?
   1. The offspring exhibit only parental combinations due to incomplete linkage
   2. The linked genes exhibit some crossing over in complete linkage
   3. The separation of two linked genes are possible in incomplete linkage
   4. Crossing over is absent in complete linkage

   a) 1 and 2 b) 2 and 3
   c) 3 and 4 d) 1 and 4

4. Due to incomplete linkage in maize, the ratio of parental and recombinants are
   a) 50:50 b) 7:1:1:7 c) 96.4: 3.6 d) 1:7:7:1

5. The point mutation sequence for transition, transition, transversion and transversion in DNA are
   a) A to T, T to A, C to G and G to C
   b) A to G, C to T, C to G and T to A
   c) C to G, A to G, T to A and G to A
   d) G to C, A to T, T to A and C to G

6. If haploid number in a cell is 18. The double monosomic and trisomic number will be
   a) 34 and 37 b) 34 and 35
   c) 37 and 35 d) 17 and 19

7. Changing the codon AGC to AGA represents
   a) missense mutation b) nonsense mutation
   c) frameshift mutation d) deletion mutation

8. **Assertion (A):** Gamma rays are generally use to induce mutation in wheat varieties.
   Reason (R): Because they carry lower energy to non-ionize electrons from atom
   a) A and R are correct
   b) A is correct. R is wrong
   c) A is wrong. R is correct
   d) A and R are wrong

9. Which one of the following pairs of codons is correctly matched with their function or the signal for the particular amino acid?
   a) UUA, UCA - Leucine
   b) GUU, GCU - Alanine
   c) UAG, UGA - Stop
   d) AUG, ACG – Start / Methionine

10. Removal of introns and joining of exons in a defined order during transcription is called
    a) Splicing b) Looping
    c) Inducing d) Slicing

11. If one strand of DNA has the nitrogenous base sequence as ATCTG, what would be the complementary RNA strand sequence?
    a) ATCGU b) TTAGU
    c) UAGAC d) AACTG

12. Removal of RNA polymerase III nucleoplasm will affect the synthesis of
    a) rRNA b) tRNA
    c) hnRNA d) mRNA

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